ripped apart, and then eaten, by the main galaxy. The Milky Way engaged in at least one act of cannibalism in the last billion years, when it consumed a dwarf galaxy whose flayed remains can be seen as a stream of stars orbiting the galactic center, beyond the stars of the constellation Sagittarius. The system is called the Sagittarius Dwarf, but should probably have been named Lunch.

In the high-density environment of clusters, two or more large galaxies routinely collide and leave behind a titanic mess: spiral structures warped beyond all recognition, newly induced bursts of star-forming regions spawned from the violent collision of gas clouds, and hundreds of millions of stars strewn hither and yon having freshly escaped the gravity of both galaxies. Some stars reassemble to form blobs that could be called dwarf galaxies. Other stars remain adrift. About ten percent of all large galaxies show evidence of a major gravitational encounter with another large galaxy—and that rate may be five times higher among galaxies in clusters.

With all this mayhem, how much galactic flotsam permeates intergalactic space, especially within clusters? Nobody knows for sure. The measurement is difficult because isolated stars are too dim to detect individually. We must rely on detecting a faint glow produced by the light of all stars combined. In fact, observations of clusters detect just such a glow between the galaxies, suggesting that there may be as many vagabond, homeless stars as there are stars within the galaxies themselves.

Adding ammo to the discussion, we have found (without looking for them) more than a dozen supernovas that exploded far away from what we presume to be their "host" galaxies. In ordinary galaxies, for every star that explodes in this way, a hundred thousand to a million do not, so isolated supernovas may betray entire populations of undetected stars. Supernovas are stars that have blown themselves to smithereens and, in the process, have temporarily (over several weeks) increased their luminosity a billion-fold, making them visible across the universe. While a dozen homeless supernovas is a relatively small number, many more may await discovery, since most supernova searches systematically monitor known galaxies and not empty space.

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There's more to clusters than their constituent galaxies and their wayward stars. Measurements made with X-ray-sensitive telescopes